

Write the answer to each question on *the question paper*. Show all working.

Question 1**(7 marks)**

a) Evaluate, correct to 2 decimal places $\sqrt{\frac{23.45^2 - 14.02^2}{6.2 \times 13.9}}$ (1)

b) Convert the recurring decimal, $2.\dot{4}\dot{5}$ to a fraction (rational number). (2)
(*show all working; without the use of the calculator*)

c) Express without using negative or fractional powers: $(y + 4)^{-\frac{2}{3}}$ (2)

d) Evaluate $\frac{a^{-3}}{a^2b^{-4}}$ when $a = \frac{1}{3}$ and $b = \frac{1}{2}$. Give answer in $\frac{a}{b}$ form. (2)

Question 2**(11 marks)**

a) Simplify:

i) $\frac{m^2 - 3mp}{3m - 9p}$ (2)

ii) $\frac{4}{x-3} - \frac{2}{x}$ (2)

iii) $\frac{x+5}{3} \div \frac{x^2-25}{15}$ (2)

b) Factorize fully:

i) $8x^2y + 20xy - 12y$ (2)

ii) $h^3 + 27$ (1)

c) Rationalize the denominator: $\frac{2\sqrt{3}}{4 - \sqrt{6}}$ (2)

Question 3**(13 marks)**

a) Solve:

i) $x^2 = 7x$ (2)

ii) $\frac{x-3}{3} - \frac{x+5}{2} = 4$ (2)

iii) $|2x-3| = -3x+7$ (3)

- b) Solve for y and graph the solution on a number line

(3)

$$|2y - 3| > 5$$

- c) Solve the pair of simultaneous equations

(3)

$$3x - 2y = 11$$

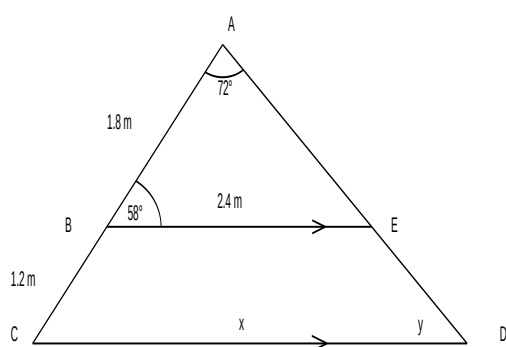
$$5x + y = 1$$

Question 4**(9 marks)**

- a) The interior angle of a regular polygon is 150° . How many sides does the polygon have?

(1)

- b) In the figure below, BE is parallel to CD .



- i) Show that $\triangle ABE$ and $\triangle ACD$ are similar triangles. Give reasons.

(2)

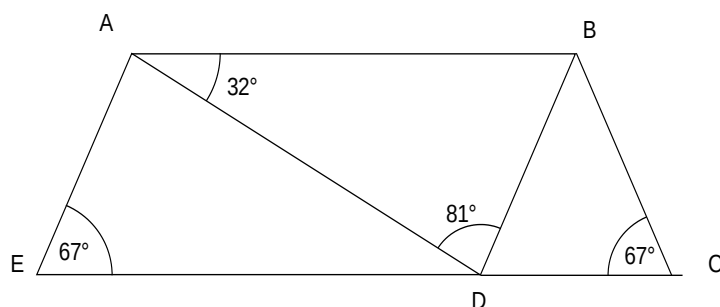
- ii) Find the value of x .

(2)

- iii) Find the value of y , giving reasons for all your working.

(1)

- c) In the diagram below, $BD = BC$, $\angle AED = \angle BCD = 67^\circ$, $\angle BAD = 32^\circ$ and $\angle ADB = 81^\circ$.



Prove, giving reasons, that $ABDE$ is a parallelogram.

(3)

Questions removed:

- a) Evaluate $(-2.7 \times 10^{-5}) \div (4.5 \times 10^8)$ writing your answer in scientific notation correct to 3 significant figures. (1)

- e) Evaluate: $-2|-5| \times |-3| + (-4)^0$ (2)

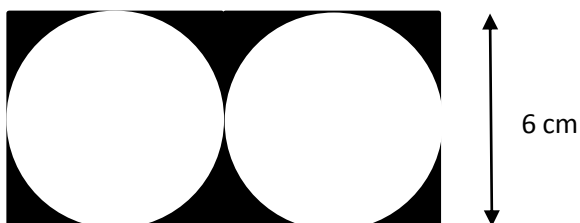
- d) Simplify $5\sqrt{27} + \sqrt{3} - 2\sqrt{12}$ leaving your answer in simplest surd form. (2)

- i) $8^{x-1} = \left(\frac{1}{2}\right)^x$ (2)

- d) Solve by completing the square. Give answer to 3 significant figures. (3)

$$x^2 - 10x + 7 = 0$$

- d) Find the area of the shaded region. Give your answer to 2 decimal places. (2)



Teacher: M Shah R Sekaran
 E Busch M Johnson
 J Meyer K Pittard

YEAR 11

Assessment 1

April 2011

2U Mathematics

Time Allowed: 50 minutes

Total Marks: 40 marks

General Instructions

- Answer **ALL** questions on the Examination paper, in the space provided.
- Write using blue or black pen **only**
- **Board approved Calculators may be used**
- Write your student number and/or name at the top of the page
- All necessary working should be shown and clearly set out in every question

Question	Mark	Out of:
Q1 – Arithmetic		7
Q2 – Algebra & Surds		11
Q3 – Equations & Inequations		13
Q4 – Geometry		9
TOTAL		40

Write the answer to each question on *the question paper*. Show all working.

Question 1

(7 marks)

- a) Evaluate, correct to 2 decimal places $\sqrt{\frac{23.45^2 - 14.02^2}{6.2 \times 13.9}}$ (1)

$$2.02$$

- b) Convert the recurring decimal, $2.\dot{4}5$ to a fraction (rational number). (2)
(show all working; without the use of the calculator)

$$\begin{aligned} \text{Let } N &= 2.4545\ldots \\ 100N &= 245.4545\ldots \\ 100N - N &= 243 \\ 99N &= 243 \\ N &= \frac{243}{99} \\ N &= 2\frac{45}{99} \\ N &= 2\frac{5}{11} \end{aligned}$$

- c) Express without using negative or fractional powers: $(y+4)^{-\frac{2}{3}}$ (2)

$$\frac{1}{\sqrt[3]{(y+4)^2}}$$

- d) Evaluate $\frac{a^{-3}}{a^2b^{-4}}$ when $a = \frac{1}{3}$ and $b = \frac{1}{2}$. Give answer in $\frac{a}{b}$ form. (2)

$$\begin{aligned} \frac{a^{-3}}{a^2b^{-4}} &= \frac{b^4}{a^5} \\ \text{Subst. } \left(\frac{1}{2}\right)^4 \div \left(\frac{1}{3}\right)^5 &= \frac{1}{16} \times \frac{243}{1} \\ &= \frac{243}{16} \end{aligned}$$

Question 2

(11 marks)

a) Simplify:

$$\begin{aligned} \text{i) } \frac{m^2 - 3mp}{3m - 9p} &= \frac{m(m - 3p)}{3(m - 3p)} \\ &= \frac{m}{3} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{ii) } \frac{4}{x-3} - \frac{2}{x} &= \frac{4x}{x(x-3)} - \frac{2(x-3)}{x(x-3)} \\ &= \frac{4x - 2x + 6}{x(x-3)} \\ &= \frac{2x + 6}{x(x-3)} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{iii) } \frac{x+5}{3} \div \frac{x^2-25}{15} &= \frac{x+5}{3} \times \frac{15}{(x+5)(x-5)} \\ &= \frac{5}{x-5} \end{aligned} \quad (2)$$

b) Factorize fully:

i) $8x^2y + 20xy - 12y$ (2)

$$= 4y(2x^2 + 5x - 3)$$

$$= 4y(2x - 1)(x + 3)$$

ii) $h^3 + 27$ (1)

$$= (h + 3)(h^2 - 3h + 9)$$

c) Rationalize the denominator: $\frac{2\sqrt{3}}{4 - \sqrt{6}}$ (2)

$$\frac{2\sqrt{3}}{4 - \sqrt{6}} \times \frac{4 + \sqrt{6}}{4 + \sqrt{6}}$$

$$= \frac{2\sqrt{3}(4 + \sqrt{6})}{16 - 6}$$

$$= \frac{8\sqrt{3} + 2\sqrt{18}}{10}$$

$$= \frac{8\sqrt{3} + 6\sqrt{2}}{10}$$

$$= \frac{4\sqrt{3} + 3\sqrt{2}}{5}$$

$$\sqrt{18} = \sqrt{9 \times 2}$$

$$= 3\sqrt{2}$$

Question 3

(13 marks)

a) Solve:

i) $x^2 = 7x$ (2)

$$x^2 - 7x = 0$$

$$x(x - 7) = 0$$

$$x = 0 \text{ or } x = 7$$

ii) $\frac{x-3}{3} - \frac{x+5}{2} = 4$ $\times 6 \text{ b.s.}$ (2)

$$2(x-3) - 3(x+5) = 4 \times 6$$

$$2x - 6 - 3x - 15 = 24$$

$$-x = 45$$

$$\underline{x = -45}$$

iii) $|2x-3| = -3x+7$ (3)

$$2x-3 = -3x+7 \quad \text{or} \quad -2x+3 = -3x+7$$

$$5x = 10$$

$$\underline{x = 2}$$

$$\underline{x = 4}$$

Test: $|4-3| = -6+7$

$$1 = 1$$



$$|8-3| = -3(4)+7$$

$$|5| = -5$$

X.

Solution is $x=2$

b) Solve for y and graph the solution on a number line

(3)

$$|2y - 3| > 5$$

$$2y - 3 > 5$$

$$2y > 8$$

$$y > 4$$

$$\text{or } 2y - 3 < -5$$

$$2y < -2$$

$$y < -1$$



c) Solve the pair of simultaneous equations

(3)

$$3x - 2y = 11 \quad \text{--- (1)}$$

$$5x + y = 1 \quad \text{--- (2)}$$

$$\begin{array}{rcl} 2 \times (2) & 10x + 2y & = 2 \\ (1) & 3x - 2y & = 11 \\ \hline & 13x & = 13 \\ & x & = 1 \end{array}$$

$$\begin{array}{rcl} (2) \rightarrow & 5x + y & = 1 \\ & 5 + y & = 1 \\ & y & = -4 \end{array}$$

$$\begin{array}{l} x = 1 \\ y = -4 \end{array}$$

Question 4

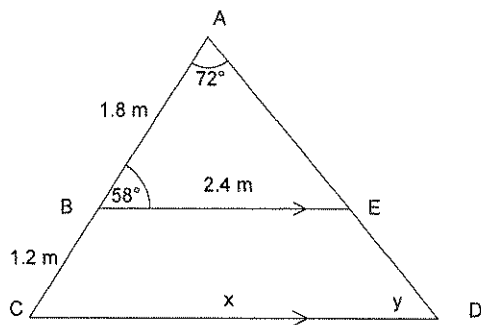
(9 marks)

- a) The interior angle of a regular polygon is 150° . How many sides does the polygon have?

$$\text{Ext. angle} = 180^\circ - 150^\circ = 30^\circ \quad (1)$$

$$\therefore \text{No of sides} = \frac{360^\circ}{30^\circ} = 12 \quad \underline{12 \text{ sides}}$$

- b) In the figure below, BE is parallel to CD .



- i) Show that $\triangle ABE$ and $\triangle ACD$ are similar triangles. Give reasons. (2)

$$\angle ABE = \angle ACD \quad (\text{corres } \angle\text{'s on } BE \parallel CD)$$

$$\angle A \text{ common to } \triangle ABE \text{ and } \triangle ACD$$

$$\therefore \triangle ABE \sim \triangle ACD \text{ similar (A.A.)}$$

- ii) Find the value of x . (2)

$$\frac{1.8}{1.8 + 1.2} = \frac{2.4}{x}$$

$$\underline{x = 4 \text{ m}}$$

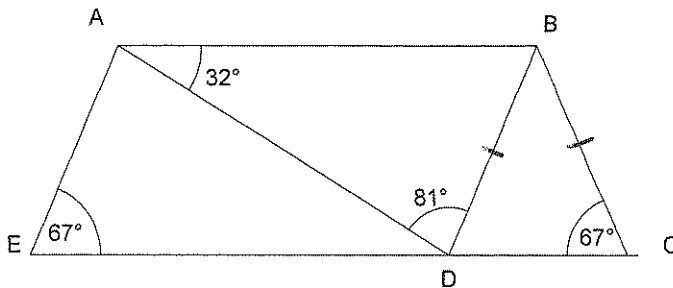
$$1.8x = 2.4 \times 3$$

- iii) Find the value of y , giving reasons for all your working. (1)

$$y = 180^\circ - (72^\circ + 58^\circ) \quad (\text{sum of } \angle \text{ in } \triangle)$$

$$= 50^\circ$$

- c) In the diagram below, $BD = BC$, $\angle AED = \angle BCD = 67^\circ$, $\angle BAD = 32^\circ$ and $\angle ADB = 81^\circ$.



Prove, giving reasons, that $ABDE$ is a parallelogram.

(3)

$$\angle BDC = 67^\circ \quad (\text{base } \angle \text{ of isosceles } \triangle BDC)$$

$$\angle AED = \angle BDC = 67^\circ$$

$$\therefore AE \parallel BD \quad (\text{corresponding } \angle \text{'s on } AE \text{ and } BD)$$

$$\begin{aligned} \angle ADE &= 180^\circ - (67^\circ + 81^\circ) \\ &= 32^\circ \quad (\text{straight } \angle) \end{aligned}$$

$$\begin{aligned} \angle BAD &= \angle ADE \quad (\text{given } \angle BAD) \\ &= 32^\circ \end{aligned}$$

$$\therefore AB \parallel DE \quad (\text{alt. } \angle \text{'s on } AB \text{ and } DE)$$

$\therefore ABDE$ is a parallelogram (two pairs of opp. sides are parallel)